

IBIT Overnight Strategy - ML Analysis

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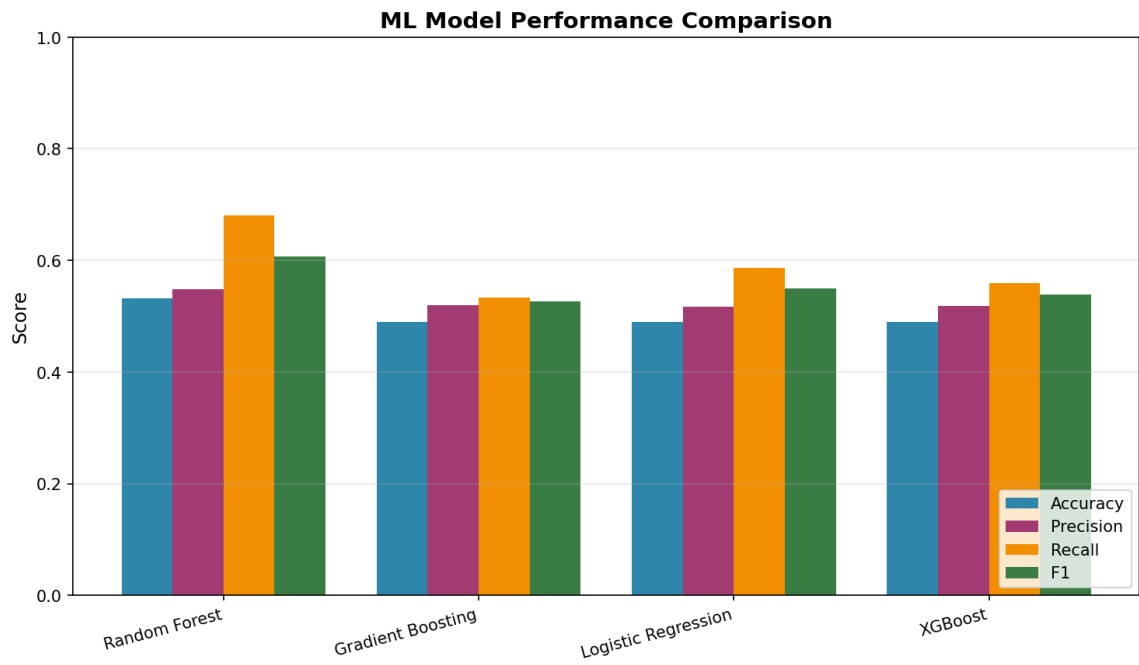
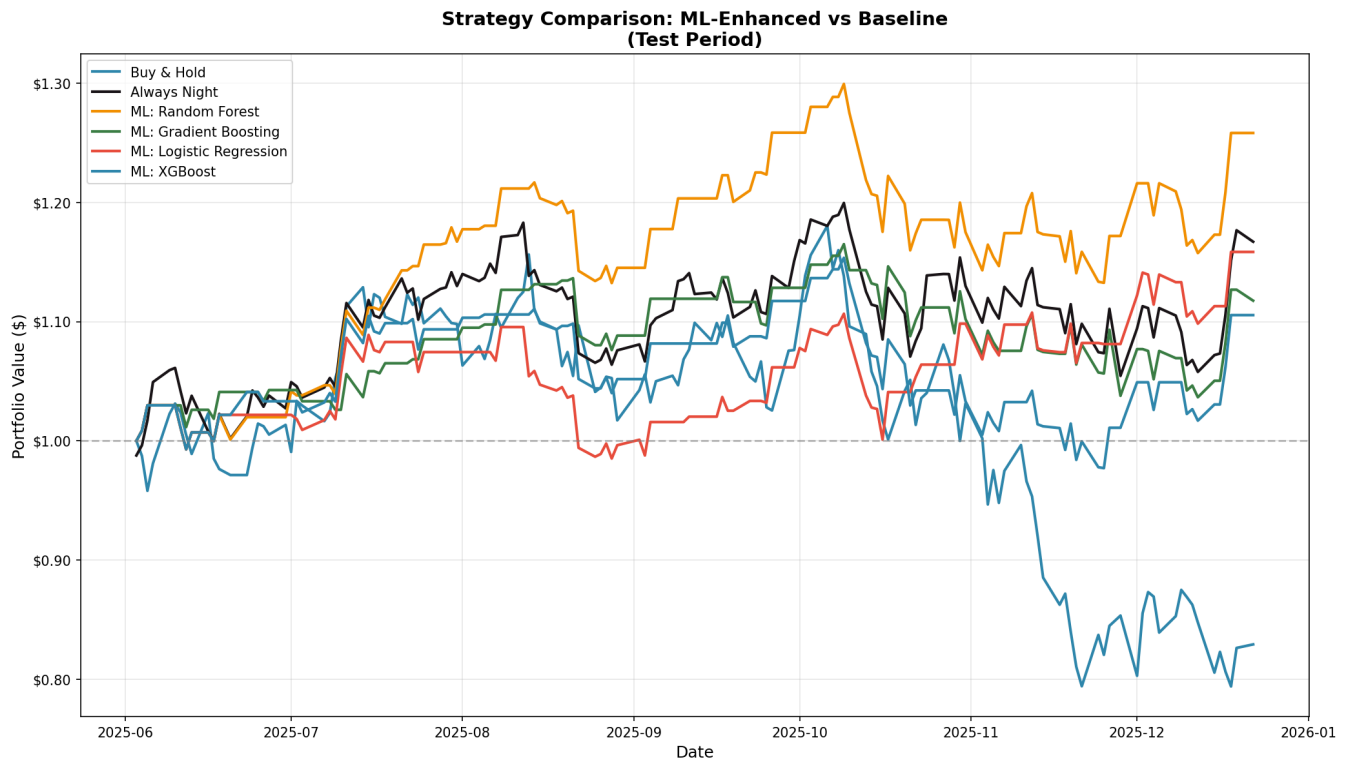
ML Model Performance

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	53.2%	54.8%	68.0%	0.61
Gradient Boosting	48.9%	51.9%	53.3%	0.53
Logistic Regression	48.9%	51.8%	58.7%	0.55
XGBoost	48.9%	51.9%	56.0%	0.54

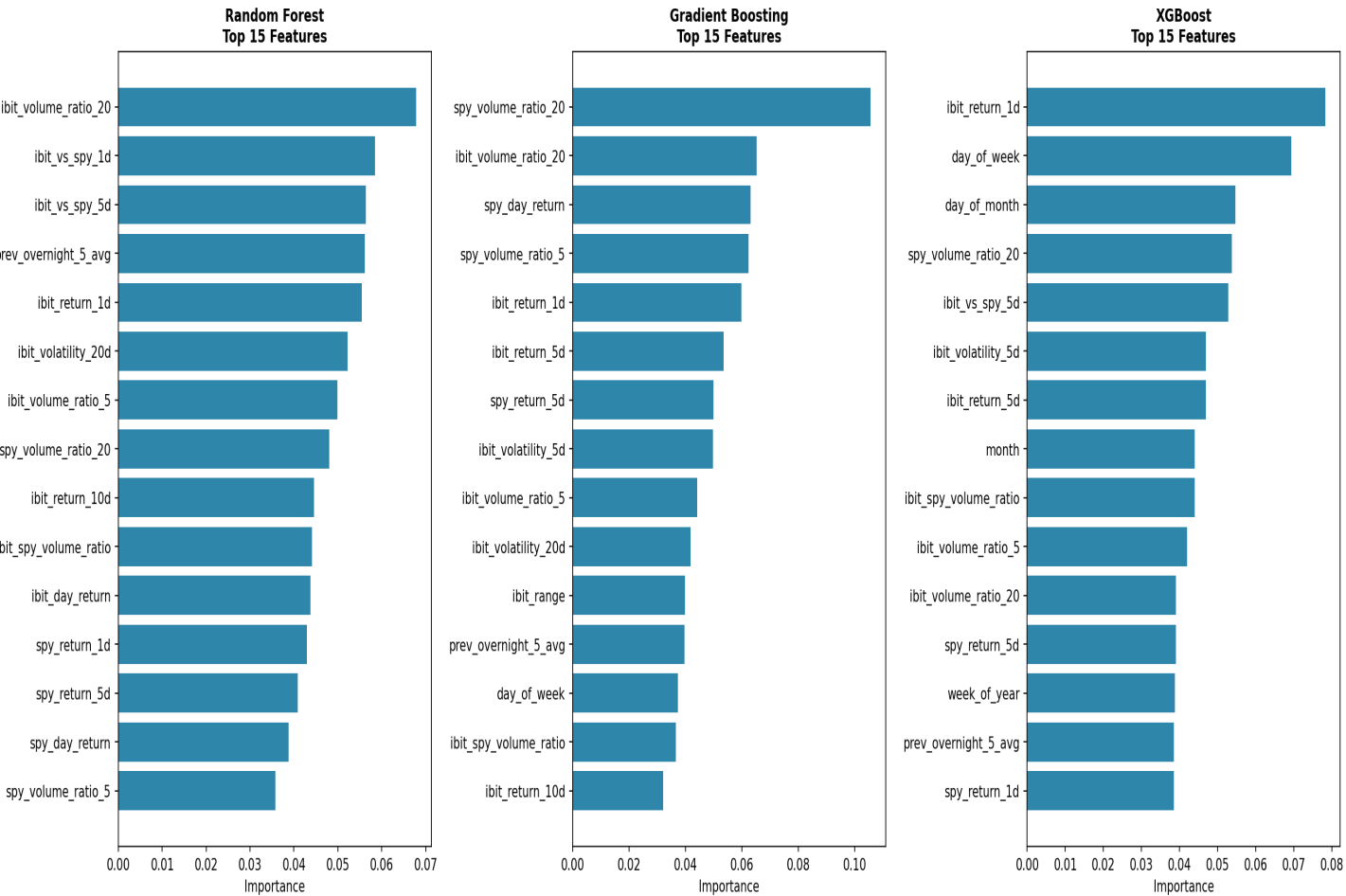
Strategy Backtest Results (Test Period)

Strategy	Total Return	Sharpe	Max DD	Win Rate	Trades
Buy & Hold	-17.1%	-0.85	-32.7%	45.4%	141
Always Night	16.7%	0.92	-12.1%	53.2%	141
ML: Random Forest	25.8%	1.84	-12.8%	54.8%	93
ML: Gradient Boosting	11.8%	0.73	-11.0%	51.9%	77
ML: Logistic Regression	15.8%	1.10	-10.1%	51.8%	85
ML: XGBoost	10.5%	0.61	-15.3%	51.9%	81

Performance Charts

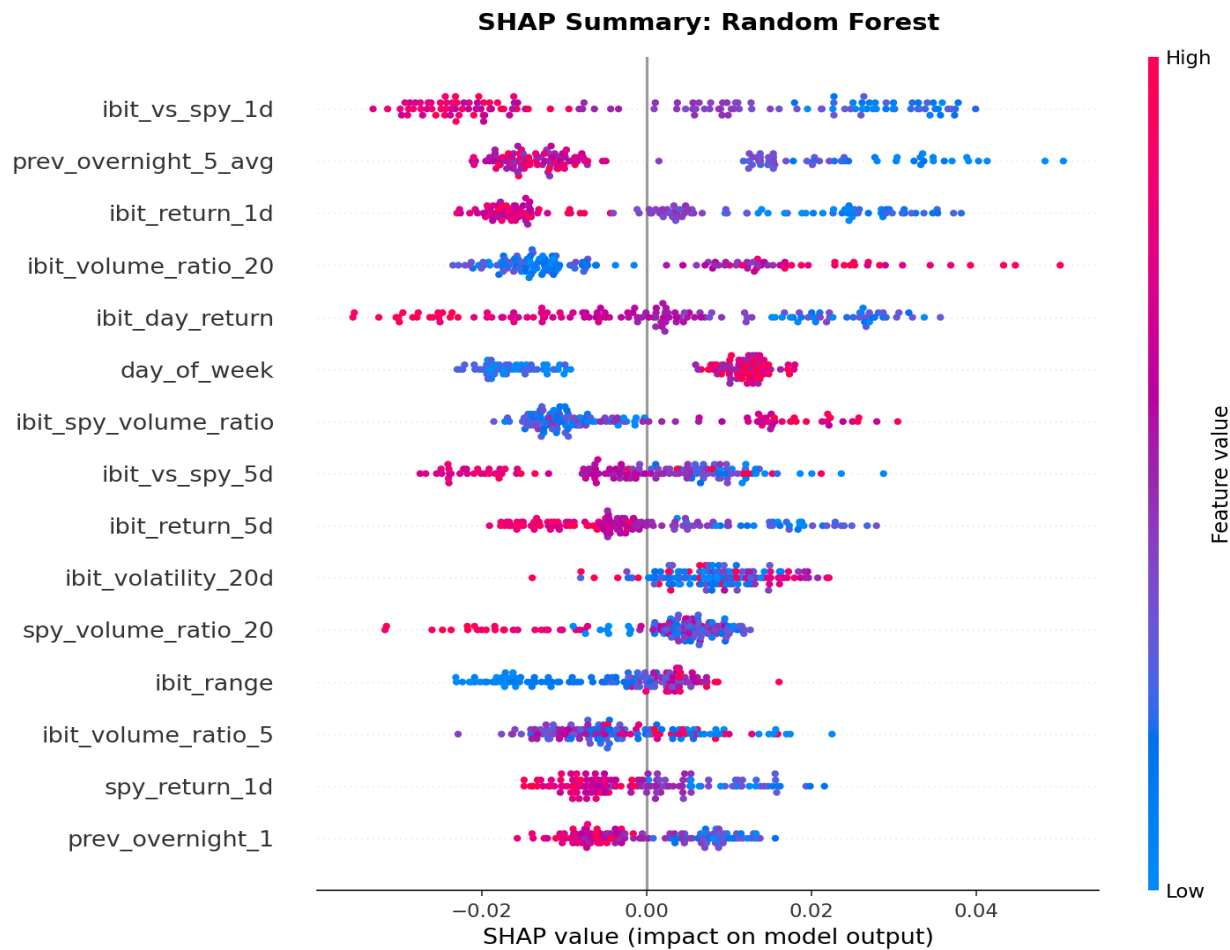


Feature Importance Analysis

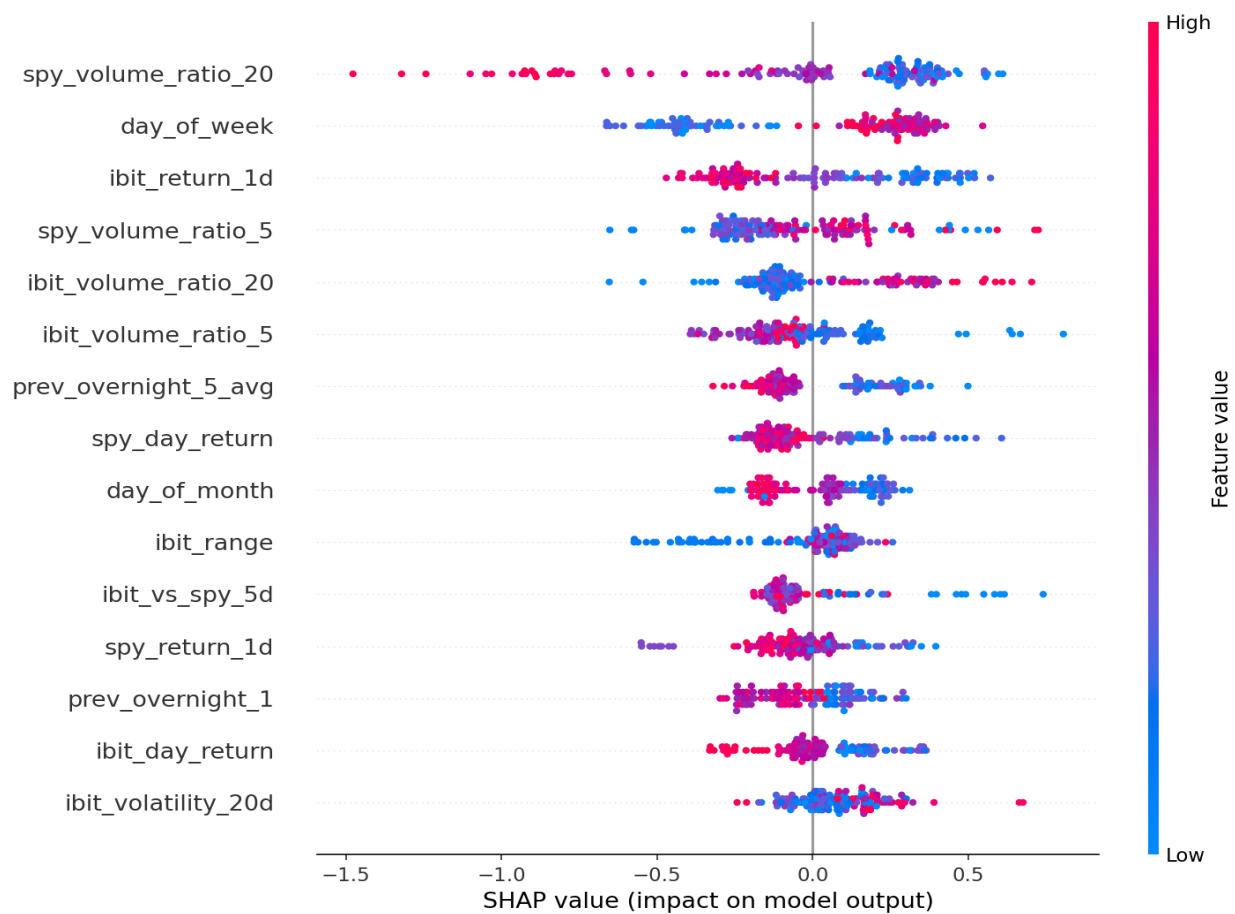


SHAP Analysis - Global Feature Importance

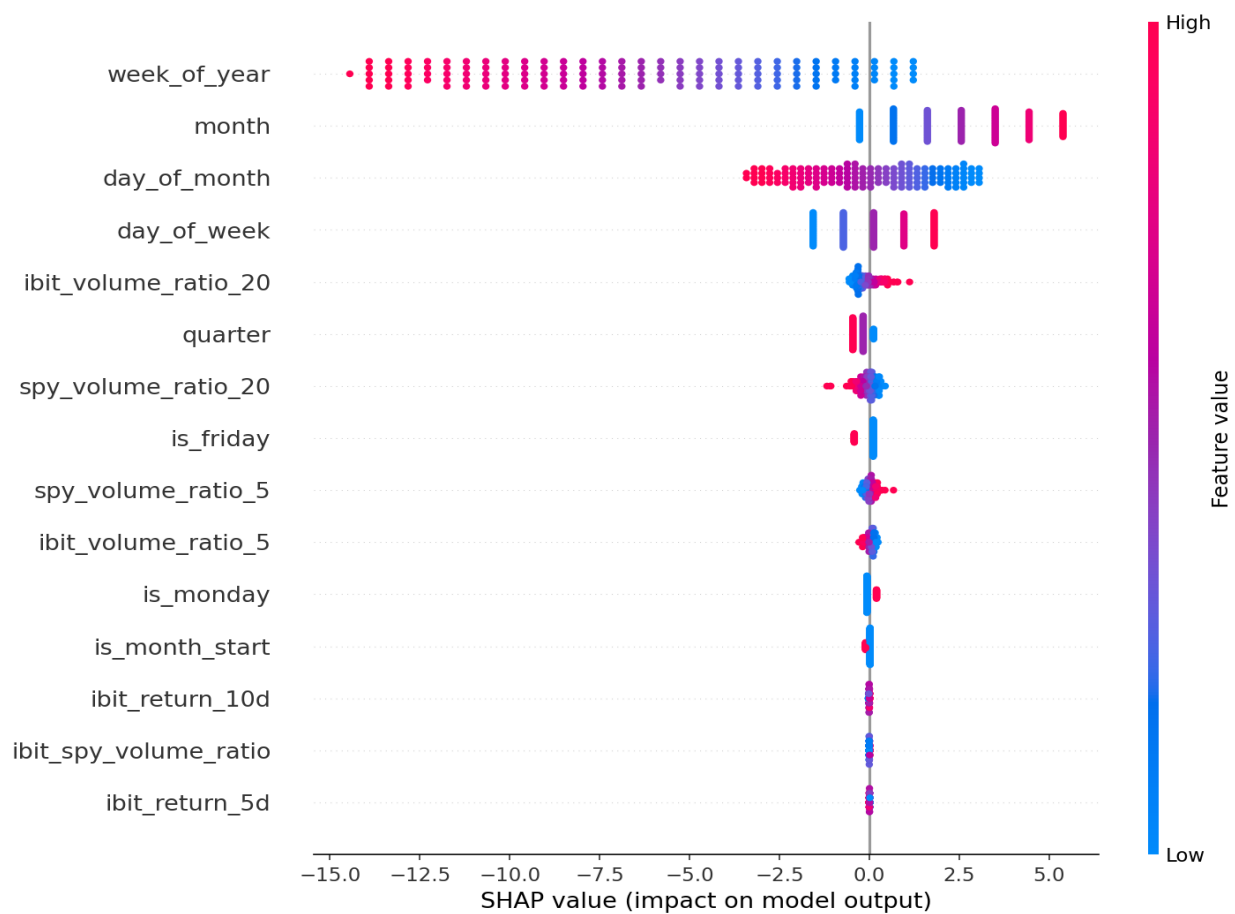
SHAP (SHapley Additive exPlanations) values quantify each feature's contribution to model predictions. Higher values indicate greater influence on the model's decision to predict positive overnight returns.



SHAP Summary: Gradient Boosting



SHAP Summary: Logistic Regression



SHAP Summary: XGBoost

